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DEVELOPMENT OF ENVIRONMENTAL SCENARIOS FOR MODELING THE FATE OF AGRICULTURAL CHEMICALS IN SOIL

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Abstract—Modeling has become a popular preliminary screening and predictive tool for assessing the environmental fate of agricultural chemicals. For modeling to be effective, realistic soil and climatic background inputs must be used. A procedure to quickly develop these environmental scenarios from readily available data sources for modeling leaching and dissipation is discussed, using the herbicide tridiphane as an example. Computerized soil survey information was used to identify soils most likely to be encountered in major use areas. A climatic model generated rainfall patterns for the study areas and provided the probability statement for the environmental assessment. Less than 5 man-days of effort were required for the formulation of four realistic scenarios to use in the evaluation of the leaching and dissipation of tridiphane in key future markets.

Keywords—Tridiphane Risk assessment Soil survey Climatic modeling

INTRODUCTION

The introduction of a new agricultural chemical requires estimation of its environmental behavior prior to actual field usage. In the past, the potential behavior of a new pesticide has been inferred from a combination of laboratory data and a few field monitoring studies. The major limitation of field studies is that each study represents only one combination of field, climatic and management factors. The researcher has little control over most of the parameters; therefore, the experiment may be subjected to extreme conditions. Modeling has become a powerful tool in many areas of environmental science to overcome the weaknesses of field studies and to allow the researcher to vary the background conditions as desired.

Modeling as a preliminary screening and predictive tool for assessing the environmental fate of pesticides is gaining increasing acceptance. The U.S. Environmental Protection Agency is developing programs to improve these predictive and risk assessment tools, especially in the areas of ground water protection [1,2]. Industry has also turned to modeling as a risk assessment tool. Union Carbide researchers [3] relied, in part, on modeling to estimate the potential for aldicarb to reach ground water supplies in Florida.

In recent years, a great deal of effort has been concentrated on the improvement of these environmental models. This interest has spawned the development of a wide assortment of models for both specific and general needs. Renard et al. [4] list 18 available hydrologic models that contain a chemical movement component for small watersheds. The development of background systems to provide the needed input information for modeling, however, has lagged behind this production of newer models. This article outlines a procedure used to develop environmental scenarios from readily available data sources for early-stage modeling of potential soil leaching and dissipation using tridiphane [2-(3,5-dichlorophenyl)-2-(2,2,2-trichloroethyl)oxirane], the active ingredient in the newly developed corn herbicide Tandem (Dow Chemical Co.), as an example.

EXPERIMENTAL METHODS

Modeling rationale

Early-stage environmental modeling should be regarded as one of the tools for estimating general trends of pesticide behavior under conditions typical of major use areas prior to the widespread usage of the chemical. To ensure that the modeling results will be truly indicative of potential field behavior, the soil and climatic scenarios must be representative of the future use areas. The creation of worst-possible-case scenarios, which have only a slight probability of actually occurring, provide little useful data for the formation of effective

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product stewardship guidelines. Also, utilization of scenarios not typical of the use areas could result in the failure to recognize potentially significant environmental pathways.

Using the general guidelines suggested by Cohen et al. [2] for identifying pesticides that may have an increased potential for contaminating ground water, tridiphane would not be rated an apparent risk because of its high adsorption coefficient (K_{oc}) of approximately 5,600 [5]. Since the preliminary review of the properties indicated that an intensive modeling assessment for tridiphane in all potential markets would not be necessary, the early-stage modeling program was designed to provide answers to very broad product stewardship questions relating to soil movement and dissipation. Since only a general assessment was needed, extensive model validation with field data and a lengthy sensitivity analysis were not considered necessary at this stage.

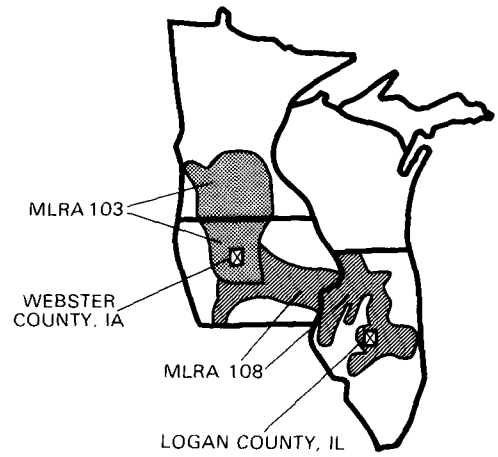

Fig. 1. Location of cornbelt study areas and counties used for weather simulations. MLRA, major land resource area.

Definition of usage areas

The major use markets for a corn herbicide such as tridiphane would be located in the cornbelt areas of the Midwest. A comparison of corn production on a county basis [6] with major land resource area (MLRA) delineations [7] indicated that the highest concentration of corn acreage occurred largely in the Central Iowa and Minnesota Till Prairie (MLRA 103), the Illinois and Iowa Deep Loess and Drift Area (MLRA 108) and the counties surrounding these two MLRAs (Fig. 1). The use of MLRAs is a convenient and effective method for dividing large geographic regions into fairly broad modeling areas containing similar patterns of soils, climate and land use.

Leaching model

The pesticide root zone model (PRZM), which has been developed and maintained by the U.S. Environmental Protection Agency, was used to estimate leaching and dissipation. The design and operation of the model are discussed in greater detail by Carsel et al. [8,9]. The model contains both a hydrology and a chemical transport component that simulate runoff, erosion, leaching, decay, plant uptake, foliar wash-off and volatilization (implicitly) of the chemical. Although PRZM had not been validated for tridiphane, the model has previously been tested with aldicarb, and the results indicate that the model can serve as an effective tool for evaluating potential ground water contaminants [8,9]. A major advantage of using PRZM is that the model has been designed so that the inputs can be obtained from commonly available data sources.

Due to computational procedures, most leaching models will generally predict the presence of some chemical in almost all soil layers. To determine the actual predicted depth of movement, the leaching profiles were terminated at the last depth that chemical residues could have been accurately determined using modern analytical techniques. The limit of determination for tridiphane in soil using electron gas chromatography is 0.05 μg/g [10].

Chemical parameters

Tridiphane is the active ingredient contained in the postemergence herbicide Tandem, which is designed to be mixed with triazine herbicides for selective grass and broadleaf control in corn. A broadcast application on 27 May at the rate of 0.84 kg/ha is typical of usage areas. In laboratory soil degradation studies, the average half-life for tridiphane was 26 d under aerobic and 3 d under anaerobic conditions, with a soil adsorption coefficient (K_{oc}) of 5,600 $\pm$ 800 [5]. Degradation rates for herbicides have been shown to decrease with increasing depth in the soil [11], although recent studies have demonstrated that appreciable numbers of microbes capable of degrading some chemicals do exist well below the surface and root zones [12]. For early-stage assessments of most pesticides, the modeler must assume degradation rates for greater soil depths based on available labora-

tory and field data that may be more representative of conditions in the plow layer. For the model simulations, the dissipation half-life in the surface or plow layer (upper 15 cm of the soil) was set equal to the laboratory degradation half-life of 26 d, and then doubled to 52 d in and below the remainder of the root zone to reflect the generally lower temperatures and less favorable microbial conditions in the subsoil. Considering the rapid degradation of tridiphane under anaerobic conditions, which would be more likely to occur periodically below the root zone, the assumed longer half-life value for deeper portions of the soil profile appears conservative.

Climatic parameters

Climatic patterns for the study areas were developed using the computer model Weather Generator (WGEN) [13]. The fundamental basis and operation of the model are described in detail by Richardson [14] and Richardson and Wright [13]. WGEN was designed to formulate synthetic weather data with the same statistical characteristics as the actual weather patterns at a given location. Outputs include daily generated values for precipitation, maximum and minimum air temperatures and solar radiation for an *n*-period at the selected locations. Webster County, Iowa, and Logan County, Illinois, were selected as the weather generation sites for MLRAs 103 and 108, respectively, because of their relatively central locations.

Modeling predictions of pesticide movement, whether in runoff or leaching, should be related to a probability of occurrence. Carsel et al. [8] suggested the development of probability statements for pesticide leaching through the utilization of climatic patterns. The probability statement produced from modeling of the leaching of tridiphane in soil was based on rainfall patterns produced by WGEN. Rather than simulate leaching for several years, a rainfall pattern with an approximate 1-in-10-year return probability was selected. To focus on the period when the highest concentrations of chemical were present, the cumulative rainfall from the day of application, 27 May to 31 July (a period encompassing at least two half-lives) was used to select the weather pattern. The entire model simulation extended to 31 December. Figure 2 is a schematic diagram of the procedure used to identify the rainfall pattern most typical of the generated patterns with a cumulative rainfall so heavy that the pattern would be expected to occur only once every 10 years, on average. Several 10-year simulations must be generated, since in any single 10-year climatic simulation, the heaviest seasonal rainfall pattern may actually have a return probability of much greater or less than 10 years.

The generated rainfall and temperature inputs used in PRZM are summarized on a monthly basis in Table 1. The total computer time required to generate the weather data was approximately 2 to 3 CPU min. The probability approach used is similar in principle to the design-frequency concept employed in the design of many structures for handling runoff waters. In designing runoff-control structures, the emphasis is on single-storm peak discharge, but for determining the leaching of a pesticide, the cumulative rainfall over a critical period of time is more important. A 1-in-10-year design frequency is commonly used in the design of grass waterways, terraces and levees for protecting farmland [15], which indicates that the use of this probability has been accepted for many aspects of environmental risk associated with normal agricultural operations.

Table 1. Monthly climatic patterns for modeled areas

Major land resource area[a]	April	May	June	July	Aug.	Sept.	Oct.	Nov.	Dec.
Total rainfall (cm)									
108	16.98	6.50	10.94	26.07	4.95	6.52	6.58	6.96	5.08
103	12.00	17.29	27.12	5.33	15.23	10.24	5.99	3.32	2.11
Average temperature (°C)[b]									
108	9.4	17.1	22.9	25.4	23.4	14.9	14.4	8.4	2.5
103	5.9	15.2	19.7	23.0	19.9	16.3	10.9	2.0	1.0

[a]For major land resource area 108, the simulation was based on Logan County, Illinois, and for area 103, the simulation was based on Webster County, Iowa. Climatic data was developed using the Weather Generator model (WGEN) [13].

[b]Average monthly air temperature were calculated from daily high and low temperatures.

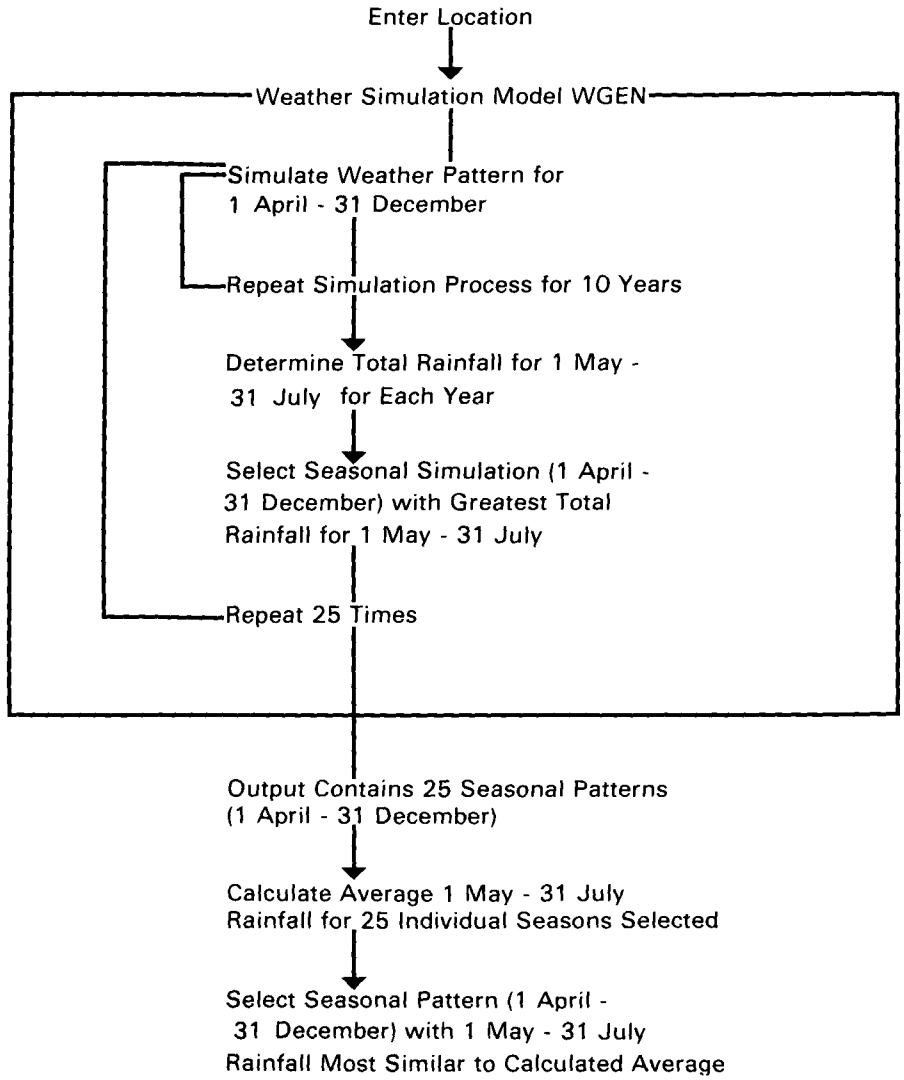

Fig. 2. Flow diagram of procedure to select weather patterns with a 1-in-10-year probability using the Weather Generator Model (WGEN) [13].

Soil parameters

To select the soils for modeling, a computerized soils library [16] containing data from the U.S. Department of Agriculture Soil Conservation Service (SCS) was used. The library consists of two separate files that can be linked for data searches. The first file contains soil data taken from the U.S. Department of Agriculture SCS mapping unit use file (MUUF) (1982 version). This file is a computerized record of the soil acreage data contained in the modern soil surveys for approximately 1,600 counties. The second file was created from the SCS Soils-5 file (1982 version) and contains physical, chemical and agronomic properties for the soil series found in the United States. The soils library has been designed to allow the information to be searched according to specific questions posed by the user.

The counties with modern soil surveys contained in the study areas served as the basis for soil selection. For each study area, the library was used to identify from the county soil survey data those soils with at least one soil phase rated for corn production in the Soils-5 file and slopes of less

than approximately 10 to 12% in the MUUF file. The acreage of each of the soils was then obtained and the corn-producing soil "most likely encountered" was identified in each study area, based on acreage data and distribution. The soils selected were the Clarion loam for MLRA 103 and the Tama silty clay loam to silt loam for MLRA 108.

Since leaching problems do not always appear in the soils most likely encountered, separate searches were also conducted for each study area to identify the common corn-producing soils with textures of sandy loam or coarser and organic matter contents of 1.5% or less in the surface horizon. The high-leaching soils selected were the Salida sandy loam for the MLRA 103 area and the Chelsea loamy fine sand for the MLRA 108 area. The acreage of the soils selected for modeling are presented in Table 2. The primary considerations for selection of the individual soils were (a) total acreage in the study area and surrounding counties, (b) number of surveyed counties containing the soil and (c) geographic distribution across the study area. Computer time required for the data searches totaled less than 5 CPU min.

The key properties of the soils used in the modeling are listed in Table 3. The soil properties were obtained from the Soils-5 data and satisfy almost all of the soil input requirements for PRZM. For soil horizons with no organic matter entry listed in the Soils-5 file, an organic matter content of 0.5% was used because of the assumed presence of corn roots. A maximum root zone of 92 cm was used

Table 2. Acreage and distribution of soils selected for modeling

Soil series[a]	MLRA	Classification	Acres[b]		Number of surveyed counties containing soil series[b]		
			Illinois and Iowa	Iowa and Minnesota	Illinois	Iowa	Minnesota
Tama	108	Typic Argiudoll	1,108,300	NA	13	18	NA
Clarion	103	Typic Hapludoll	NA	1,494,400	NA	19	10
Chelsea	108	Alfic Udipsamment	71,600	NA	4	21	NA
Salida	103	Entic Hapludoll	NA	35,700	NA	15	9

MLRA, major land resource area; NA, not applicable (the area was not considered part of the study area for the particular soil).
[a]All official variants are included for each soil.
[b]Based on data obtained from U.S. Department of Agriculture Soil Conservation Service mapping unit use file (1982 version), including counties within and surrounding the particular MLRA.

Table 3. Key soil properties for soils modeled[a]

Soil	Horizon depth[b] (cm)	Texture[c]	% sand	% clay	Bulk density (g/cm^3)	% organic matter	Hydrologic group
Tama	0–36	SICL-SIL	3.0	27.0	1.25	3.50	B
	36–114	SICL	3.0	31.0	1.33	0.50	
	114–160	SICL	3.0	27.0	1.37	0.50	
Clarion	0–43	L	37.5	21.0	1.42	3.50	B
	43–81	L-CL	37.5	27.0	1.60	0.50	
	81–160	L	42.5	17.0	1.75	0.50	
Chelsea	0–15	LFS	77.5	11.5	1.52	0.75	A
	15–180	FS	91.0	7.5	1.62	0.50	
Salida	0–15	SL	78.0	10.0	1.40	0.75	A
	15–30	LS	95.0	5.0	1.58	0.50	
	30–160	S	97.5	2.5	1.58	0.50	

[a]Information obtained from U.S. Department of Agriculture Soil Conservation Service Soils-5 interpretation file (1982 version).
[b]For actual model simulation, a 15-cm plow layer was separated from the surface horizon if necessary.
[c]SICL, silty clay loam; SIL, silt loam; L, loam; LFS, loamy fine sand; FS, fine sand; SL, sandy loam; LS, loamy sand; S, sand.

Table 4. Calculated leaching profiles for tridiphane following a 0.84-kg/ha broadcast application on 27 May[a]

Soil	Soil depth[b] (cm)	Total soil concentration (μg/g)							
		31 May	30 June	31 July	31 Aug.	30 Sept.	31 Oct.	30 Nov.	31 Dec.
Tama	0–5	0.94	0.42	0.18	0.08	NC	NC	NC	NC
	5–10	NC	NC	NC	NC	NC	NC	NC	NC
	10–160	NC	NC	NC	NC	NC	NC	NC	NC
Clarion	0–5	0.82	0.36	0.16	0.07	NC	NC	NC	NC
	5–10	NC	NC	NC	NC	NC	NC	NC	NC
	10–160	NC	NC	NC	NC	NC	NC	NC	NC
Chelsea	0–5	0.77	0.33	0.12	0.05	NC	NC	NC	NC
	5–10	NC	NC	NC	NC	NC	NC	NC	NC
	10–180	NC	NC	NC	NC	NC	NC	NC	NC
Salida	0–5	0.80	0.30	0.13	0.05	NC	NC	NC	NC
	5–10	NC	0.07	NC	NC	NC	NC	NC	NC
	10–160	NC	NC	NC	NC	NC	NC	NC	NC

NC, none calculated (<0.05 μg/g).
[a]Calculated using the pesticide root zone model, with 1-in-10-year seasonal rainfall probability.
[b]Soil concentration data for soil depths below 10 cm are combined for presentation. Calculations were completed for each individual 5-cm zone to the bottom of the soil profile.

for corn. The soil horizons were differentiated according to the Soils-5 data, with the addition of an assumed 15-cm surface plow layer.

Soil surveys are the most accurate and comprehensive source of nationwide soils information available for predictive modeling of widely used agricultural chemicals. However, the limitations of this soils information need to be recognized by the user. Due to numerous interpretative and cartographic limitations, a certain amount of variability exists in the basic soil mapping units [17]. In addition, nontypical site characteristics, such as animal burrows, may alter important modeling parameters for an individual field. Due to the possible variation in soils, the soil survey data should be used as reasonable estimates of typical conditions for the purpose of modeling and not as a reference for the exact conditions at any particular site [16].

Management parameters

To emphasize the potential leaching aspect, soil management inputs were designed to maximize infiltration. Continuous corn operations using conservation tillage on a field with a length of 91 m and a slope of 1% were assumed. A crop emergence date of 10 May and a final maximum canopy cover of 90% were considered typical for the study areas.

MODELING RESULTS

The predicted soil concentration profiles for tridiphane in each of the four soils modeled are summarized in Table 4. Dissipation of the tridiphane was fairly rapid, and by the end of 127 d of simulation, tridiphane was not detected in any soil. The rapid soil degradation and high soil adsorption limited the predicted movement of tridiphane to the upper 10 cm of all soil profiles studied, including the two coarse-textured, low-organic matter soils.

CONCLUSIONS

Realistic environmental scenarios for the modeling of agricultural chemicals can be developed quickly with the use of computerized soil databases and a weather model. Less than 5 man-days of effort were required for the formulation of the four scenarios used to evaluate the leaching and dissipation of tridiphane in important corn markets. This study involved the use of the model PRZM, but the scenario development techniques would be equally effective in providing the scenario inputs for many other physically based environmental models not dependent on site-specific calibration parameters. In addition to providing the needed modeling input information, these sources of background information also offer effective methods for assigning probability factors to the risk assessment.

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